King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 241

Original exam date: 2 October 2024. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $\int \frac{\sec x \tan x}{\sec x - 1}\, dx =$

(a) $\ln|\sec x + 1| + c$

(b) $\ln|\tan x + 1| + c$

(c) $\ln|\sec x - 1| + c$

(d) $\ln|\sec x + \tan x| + c$

(e) $\ln|\tan x - 1| + c$

2. $\int \frac{x(x-2)}{(x-1)^3}\, dx =$

(a) $\ln|x-1| + \dfrac{1}{2(x-1)^2} + c$

(b) $\ln|x-1| + \dfrac{1}{(x-1)^2} + c$

(c) $\ln|x| + \dfrac{1}{(x-1)^2} + c$

(d) $\ln|x| + \dfrac{1}{2(x-1)^2} + c$

(e) $\ln|x-1| + \dfrac{2}{(x-1)^2} + c$

3. The area of the region bounded by the graphs of the equations $y = -x^2 + 3x + 1$ and $y = -x + 1$ is

(a) $\dfrac{32}{3}$

(b) 11

(c) $\dfrac{31}{3}$

(d) $\dfrac{34}{3}$

(e) $\dfrac{35}{3}$

4. If $F(x) = \int_2^{x^2} \frac{1}{t^3}\, dt$, then $F'(2) =$

(a) $\frac{1}{8}$

(b) $\frac{1}{16}$

(c) $\frac{1}{32}$

(d) $\frac{1}{2}$

(e) $\frac{1}{4}$

5. Use the left end points and 3 rectangles to find an approximation of the area of the region between the graph of the function $f(x) = 2x^2 - x - 1$ and the x-axis over the interval $[2, 5]$.

(a) 45

(b) 49

(c) 47

(d) 46

(e) 51

6. $\int_{-\pi/2}^{\pi/2} (\sin 4x + \cos 4x)\, dx =$

(a) 1

(b) -2

(c) 0

(d) -1

(e) 2

7. The lower sum (in terms of n, the number of subintervals), for the region bounded by the graph of the function $f(x) = 9 - x^2$ on the interval $[0, 2]$ is:

(a) $\frac{64}{3} - \frac{4}{n} + \frac{4}{3n^2}$

(b) $\frac{46}{3} - \frac{4}{n} - \frac{4}{3n^2}$

(c) $\frac{46}{3} - \frac{4}{n} + \frac{4}{3n^2}$

(d) $\frac{46}{3} + \frac{4}{n} + \frac{4}{3n^2}$

(e) $\frac{46}{3} + \frac{4}{n} - \frac{4}{3n^2}$

8. $\int_0^2 \frac{x\,dx}{\sqrt{1+2x^2}} =$

(a) 2

(b) 4

(c) 1

(d) 3

(e) 5

9. Possible values of a and b that make the statement $\int_a^b \cos x\,dx = 0$ true, are:

(a) $a = \dfrac{\pi}{2},\ b = 2\pi$

(b) $a = 0,\ b = \dfrac{\pi}{2}$

(c) $a = \dfrac{\pi}{2},\ b = \dfrac{3\pi}{2}$

(d) $a = \pi,\ b = 2\pi$

(e) $a = 0,\ b = \dfrac{3\pi}{2}$

10. $\int_0^{\sqrt[3]{\ln 2}} x^2 \operatorname{sech}^2 x^3\,dx =$

(a) $\dfrac{3}{5}$

(b) $\dfrac{6}{5}$

(c) $\dfrac{2}{5}$

(d) $\dfrac{1}{5}$

(e) $\dfrac{7}{5}$

11. $\int_0^2 |2x-1|\,dx =$

(a) $\dfrac{7}{2}$

(b) $\dfrac{11}{2}$

(c) $\dfrac{5}{2}$

(d) $\dfrac{9}{2}$

(e) $\dfrac{3}{2}$

12. $\int_{-2}^{3} \frac{dx}{x^2+4x+8} =$

(a) $\arctan \dfrac{3}{2}$

(b) $\dfrac{1}{2} \arctan \dfrac{3}{2}$

(c) $2 \arctan \dfrac{2}{5}$

(d) $\arctan \dfrac{5}{2}$

(e) $\dfrac{1}{2} \arctan \dfrac{5}{2}$

13. $\int_{1}^{3} 4\,dx + \int_{-2}^{2} \sqrt{4-x^2}\,dx =$

(a) $8+\pi$

(b) $4+\pi$

(c) $8+2\pi$

(d) $8-\pi$

(e) $4+2\pi$

14. On the interval $[0,\pi]$, $\lim_{\|\Delta\|\to 0} \sum_{i=1}^{n} (2^{-c_i} \sin c_i)\,\Delta x_i$, where c_i is any point in the ith subinterval, can be written as:

(a) $\int_{0}^{2^{-\pi}} \sin x\,dx$

(b) $\int_{0}^{\pi} 2^{-x} \sin x\,dx$

(c) $\int_{0}^{\pi} 2^{x} \sin x\,dx$

(d) $\int_{0}^{\pi} 2^{-x} \cos x\,dx$

(e) $\int_{\pi}^{0} 2^{-x} \sin x\,dx$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$u = \sec x - 1$, $du = \sec x \tan x\, dx$: $\ln\lvert \sec x - 1\rvert + c$.
2	**(a)**	$x(x-2) = (x-1)^2 - 1$: $\int \left[\frac{1}{x-1} - \frac{1}{(x-1)^3}\right] dx$.
3	**(a)**	Intersect at $x = 0, 4$: $\int_0^4 (4x - x^2)\, dx = \frac{32}{3}$.
4	**(b)**	$F'(x) = \frac{1}{(x^2)^3} \cdot 2x = \frac{2}{x^5}$, so $F'(2) = \frac{1}{16}$.
5	**(d)**	$\Delta x = 1$, endpoints $2, 3, 4$: $5 + 14 + 27 = 46$.
6	**(c)**	$\sin 4x$ is odd (integral 0); $\frac{1}{4}\sin 4x$ from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ is 0.
7	**(b)**	Decreasing: right endpoints; $18 - \frac{4(n+1)(2n+1)}{3n^2} = \frac{46}{3} - \frac{4}{n} - \frac{4}{3n^2}$.
8	**(c)**	$u = 1 + 2x^2$: $\frac{1}{4}\int_1^9 u^{-1/2} du = \frac{1}{4}(6 - 2) = 1$.
9	**(d)**	$\sin b - \sin a = 0$ only for $a = \pi$, $b = 2\pi$ among the pairs.
10	**(d)**	$u = x^3$: $\frac{1}{3}\tanh(\ln 2) = \frac{1}{3} \cdot \frac{3}{5} = \frac{1}{5}$.
11	**(c)**	Split at $x = \frac{1}{2}$: $\frac{1}{4} + \frac{9}{4} = \frac{5}{2}$.
12	**(e)**	$(x+2)^2 + 4$, $u = x + 2$: $\frac{1}{2}\arctan\frac{u}{2}$ from 0 to 5 gives $\frac{1}{2}\arctan\frac{5}{2}$.
13	**(c)**	Rectangle area 8 plus semicircle area 2π: $8 + 2\pi$.
14	**(b)**	Riemann sum of $f(x) = 2^{-x}\sin x$ on $[0, \pi]$ gives $\int_0^\pi 2^{-x}\sin x\, dx$.